

THE GENERALIZED EXACT OVERLAPS CONJECTURE ON THE DIMENSION OF SELF-SIMILAR MEASURES ON THE LINE

GPT-6 ASTRA

ABSTRACT. We give a candidate proof of the generalized exact overlaps conjecture for finite self-similar measures on the line, allowing unequal signed contraction ratios and exact overlaps. Poisson partitions relate the entropy lost under addition to an integrated local variance. The same partitions detect this variance through a Gaussian channel when the stationary measure has dimension less than one. Saturation of prefix information and a final passage to iterated systems yield the proposed entropy-rate dimension formula. The argument uses exact dimensionality but no arithmetic separation hypothesis.

This model output contributed to the development of [KK26]. Prompting was done by C.K. using the prompter's local context. References and attributions were revised on 5 October 2026.

1. INTRODUCTION

Let $\Phi = \{f_i(x) = r_i x + t_i : 1 \leq i \leq m\}$ be a finite family of similarities of $\mathbb{R}$, where $0 < |r_i| < 1$, and let $p = (p_1, \dots, p_m)$ be a positive probability vector. The corresponding self-similar measure is the unique probability measure satisfying

$$\nu = \sum_{i=1}^m p_i (f_i)_* \nu. \quad (1.1)$$

Existence, uniqueness and compactness of the support are classical; see Hutchinson [Hut81]. Feng and Hu [FH09] proved that ν is exact dimensional: there is $\alpha \in [0, 1]$ such that

$$\lim_{r \downarrow 0} \frac{\log \nu([x - r, x + r])}{\log r} = \alpha \quad \text{for } \nu\text{-almost every } x.$$

We write $\dim \nu = \alpha$. All logarithms in this paper are natural.

Put $\mu = \sum_i p_i \delta_{f_i}$, viewed as a probability measure on the group of invertible affine maps of $\mathbb{R}$, and let

$$H(p) = - \sum_i p_i \log p_i, \quad \chi_\mu = \sum_i p_i \log |r_i|^{-1}.$$

For independent letters $\omega_1, \omega_2, \dots$ with law p , write

$$f_{\omega_1} \circ \dots \circ f_{\omega_n}(x) = R_n x + B_n, \quad h_\mu = \lim_{n \rightarrow \infty} \frac{H(R_n, B_n)}{n}. \quad (1.2)$$

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Here H denotes Shannon entropy. The limit exists and equals the infimum, since the entropy of the product of two independent random maps is at most the sum of their entropies. Thus h_μ is the entropy rate of the similarity-valued random walk.

An *exact overlap* is an equality between the maps defined by two distinct words of the same length. In the absence of exact overlaps, $H(R_n, B_n) = nH(p)$ and consequently $h_\mu = H(p)$. In general, the usual coding argument, applied after merging equal word maps, gives

$$\dim \nu \leq \min \left\{ 1, \frac{h_\mu}{\chi_\mu} \right\}. \quad (1.3)$$

The generalized exact overlaps conjecture asserts equality in (1.3); see Conjecture 3 in Varjú [Var26].

Hochman's theorem [Hoc14] gives the expected dimension under exponential separation and settles algebraic systems without exact overlaps. Breuillard and Varjú obtained algebraic approximation results for dimension drop [BV19] and entropy bounds in terms of Mahler measure [BV20]. Varjú [Var19] proved that Bernoulli convolutions with transcendental parameter in $(1/2, 1)$ have dimension one.

Rapaport [Rap22] settled systems with algebraic contraction ratios and arbitrary translations, without exact overlaps. Rapaport and Varjú [RV24] proved further cases for homogeneous three-map systems. Their Appendix A treats rational translations and an arbitrary positive common contraction, without exact overlaps. Feng and Feng [FF25] treated algebraic translations with an arbitrary nonzero real common contraction, again without exact overlaps.

Super-exponential concentration need not imply exact overlaps [Bak21] and [BK21]. Kittle and Kogler [KK25] allow exact overlaps and contraction on average under weakened separation hypotheses, building on the variance-summation method of Kittle [Kit25]. We also use a conditional form of cumulative paired entropy [KMD16], for which the required estimate is proved below. The following is the main claim of this manuscript.

Theorem 1.1. *Let ν be the self-similar measure associated to Φ and p as above. Then*

$$\dim \nu = \min \left\{ 1, \frac{h_\mu}{\chi_\mu} \right\}.$$

The contraction ratios may have either sign, and exact overlaps are allowed.

Corollary 1.2. *If Φ has no exact overlaps, then*

$$\dim \nu = \min \left\{ 1, \frac{H(p)}{\chi_\mu} \right\}.$$

In particular, a Bernoulli convolution with a transcendental parameter $\lambda \in (1/2, 1)$ has dimension one.

Proof. The first assertion follows from $h_\mu = H(p)$. For the second, an equality between two distinct finite words would give a nonzero polynomial with coefficients in $\{-1, 0, 1\}$ vanishing at λ . Thus there are no exact overlaps, while $H(p) = \log 2 > \log \lambda^{-1} = \chi_\mu$. $\square$

Ideas of the proof. For a finite law ρ and a scale r , reveal the cell of a sample in a uniformly shifted grid of mesh r . Write $V_\rho(r)$ for the average conditional variance divided by r^2 . This normalized local variance measures the uncertainty remaining within a cell, relative to its size. To add its contributions over scales, write

$$\mathcal{E}_R(\rho) = \int_0^R V_\rho(r) \frac{dr}{r}, \quad \mathcal{E} = \mathcal{E}_\infty.$$

The measure dr/r gives equal weight to each doubling of scale. Section 2 gives the precise definition. Two estimates connect this accumulated variance to entropy.

First, for independent finite variables,

$$\sum_{i=1}^k H(Y_i) - H(Y_1 + \cdots + Y_k) \leq 108k \sum_{i=1}^k \mathcal{E}(Y_i). \quad (1.4)$$

Reveal each input by independent Poisson cuts. Each cut contributes a conditional binary entropy; a dispersion estimate bounds their total by the right-hand side. For a sum of L independent digits with at most M values each, the sum of k copies has entropy at most $L(M-1)\log(k+1)$, since each digit sum is determined by its multiplicities. Applying (1.4) inside cells of length R therefore bounds the entropy below scale R by $\mathcal{E}_R$ and this count term.

Second, assume $\alpha = \dim \nu < 1$. Self-similarity gives a uniform lower bound for $t \|(\nu * \gamma_t)'\|_1$, where γ_t is the centered Gaussian density of standard deviation t . This makes the two ordered child laws at a cut distinguishable after Gaussian smoothing; their information contribution controls pqd^2/r^2 , where p, q are the child probabilities, d their mean gap, and r the parent length. The expected sum over Poisson cuts with parent length at most R equals $\mathcal{E}_R$ plus a nonnegative boundary term. Thus the same partitions bound entropy lost under addition and detect fine variance.

There is little information left to detect in a long prefix. Let W_n be the first n letters and X their canonically coded stationary variable. With Z standard normal, the chain rule and the law of large numbers give

$$I(W_n; X + tZ) \leq n\alpha\chi_\mu \quad (t > 0), \quad I(W_n; X + e^{-bn}Z) = n\alpha\chi_\mu + o(n) \quad (b > \chi_\mu).$$

Conditioning on R_n leaves an independent background $R_n X'$ with the same Gaussian roughness. The detection estimate consequently yields $\mathbb{E}_{R_n} \mathcal{E}_{e^{-bn}}(B_n | R_n) = o(n)$.

To apply the finite-product bound, condition on each letter's ratio. The translations become independent at a cost of $nH(R_1)$. Concavity transfers their variance to the laws $B_n | R_n$, while adding the stationary tail bounds their Gaussian entropy by that of ν . At $R = e^{-bn}$, letting $n \rightarrow \infty$, then $k \rightarrow \infty$ and $b \downarrow \chi_\mu$, gives

$$h_\mu \leq H(R_1) + \alpha\chi_\mu.$$

Apply this bound to the N -fold system and divide by N . Its entropy rate and Lyapunov exponent are Nh_μ and $N\chi_\mu$, whereas $H(R_N) \leq m \log(N+1)$. Letting $N \rightarrow \infty$ removes the ratio cost. The coding upper bound completes the argument.

Sections 2–4 prove the analytic estimates; Sections 5–7 apply them to word maps. The appendix supplies the elementary entropy and Gaussian facts. Exact dimensionality is the substantial external input.

2. INFORMATION LOST UNDER ADDITION

We call a random variable *finite* if its support is finite. We use H for discrete entropy, h for differential entropy when it is defined, and I for mutual information. A functional of a law is also written as a functional of a random variable with that law. All added Gaussian noises are independent of the inputs. Conditional laws on events of probability zero are omitted from sums.

We first describe the variance at one scale. For a bounded probability law ρ and an interval I , write

$$F_\rho(I) = \rho(I) \operatorname{Var}(\rho \mid I) = \inf_{c \in \mathbb{R}} \int_I (x - c)^2 d\rho(x).$$

Thus $F_\rho(I)$ is the squared error of replacing the points in I by their conditional mean, weighted by the mass of I . Zero-mass intervals contribute zero.

Let $I(a, r) = [a, a + r)$. For a grid of mesh r and origin s , revealing the cell of a sample leaves average conditional variance $\sum_{j \in \mathbb{Z}} F_\rho(I(s + jr, r))$. Average over $s \in [0, r)$ and divide by r^2 to define the *normalized local variance*

$$V_\rho(r) = \frac{1}{r^3} \int_{\mathbb{R}} F_\rho(I(a, r)) da. \quad (2.1)$$

The factor r^{-3} therefore consists of r^{-1} for averaging the grid origin and r^{-2} for normalizing the squared error. We also write $V_Y(r)$ when Y has law ρ .

For example, if $\rho = p\delta_0 + q\delta_d$, where $p + q = 1$ and $d > 0$, then

$$V_\rho(r) = pq \left(\frac{d}{r} \right)^2 \left(1 - \frac{d}{r} \right)_+.$$

Indeed, only intervals containing both atoms contribute; their origins form a set of length $(r - d)_+$, and their weighted variance is pqd^2 . Thus sufficiently fine cells distinguish the atoms, whereas at very coarse scales their separation is small relative to the cell size.

The proof will accumulate these local contributions over logarithmic scales. For convenience, define their integral below R by

$$\mathcal{E}_R(\rho) = \int_0^R V_\rho(r) \frac{dr}{r}, \quad \mathcal{E}(\rho) = \mathcal{E}_\infty(\rho). \quad (2.2)$$

We call this the variance energy. The logarithmic measure dr/r assigns the same weight to every interval of scales $[r, 2r]$.

Lemma 2.1. *The functionals $F_\rho(I)$, $V_\rho(r)$ and $\mathcal{E}_R(\rho)$ are concave in ρ . Moreover, $0 \leq V_\rho(r) \leq 1/4$ and, for $a \neq 0$,*

$$V_{aY+b}(r) = V_Y(r/|a|), \quad \mathcal{E}(aY + b) = \mathcal{E}(Y).$$

If ρ is finite and its support has diameter at most D , then $\mathcal{E}(\rho) < \infty$ and

$$\mathcal{E}(\rho) - \mathcal{E}_R(\rho) \leq \frac{D^2}{8R^2}. \quad (2.3)$$

Proof. The infimum defining $F_\rho(I)$ is taken over linear functionals of ρ , so $F_\rho(I)$ is concave; averaging and integration give the same property for $V_\rho(r)$ and $\mathcal{E}_R$. Each conditional law in a cell of length r has variance at most $r^2/4$. This follows by estimating squared error using the cell midpoint. The grid interpretation therefore gives $V_\rho(r) \leq 1/4$.

An affine change of variables scales cell lengths by $|a|$ and variances by a^2 , proving the identity for V . Substitution in (2.2) gives the identity for $\mathcal{E}$; interval endpoints do not affect the average over origins, also when $a < 0$.

If the support has diameter at most D , the same midpoint estimate gives $\text{Var}(\rho \mid I) \leq D^2/4$. Since $\int_{\mathbb{R}} \rho(I(a, r)) da = r$, we have $V_\rho(r) \leq D^2/(4r^2)$. Hence

$$\mathcal{E}(\rho) - \mathcal{E}_R(\rho) \leq \frac{D^2}{4} \int_R^\infty \frac{dr}{r^3} = \frac{D^2}{8R^2}.$$

For a finite law $V_\rho(r)$ also vanishes below the smallest positive atom spacing. Thus the full energy is finite. A point mass has zero energy. $\square$

We next estimate the information revealed by a single cut. The natural intermediate quantity is the mean absolute distance between independent samples:

$$\mathcal{D}(X) = \mathbb{E}|X - X'|, \quad X' \text{ an independent copy of } X.$$

For finite X, S , define the conditional cumulative entropy

$$\mathcal{A}(X \mid S) = \int_{\mathbb{R}} H(\mathbf{1}_{\{X \leq a\}} \mid S) da.$$

This is a conditional version of cumulative paired Shannon entropy; see [KMD16] for that functional. It adds, over all cut locations a , the uncertainty in which side contains X , after S is known. Lemma 2.2 controls this uncertainty by $\mathcal{D}$; Lemma 2.3 then converts the accumulated dispersion into normalized local variance.

Lemma 2.2 (Conditional cumulative entropy). *If X, Y are independent finite real variables, then*

$$\mathcal{A}(X \mid X + Y) \leq 9(\mathcal{D}(X) + \mathcal{D}(Y)). \quad (2.4)$$

Proof. We begin with a quantile estimate. Suppose $Z \in [0, 1]$ stochastically dominates a uniform variable, and let $T = \mathbb{E}(1 - Z)$. Then

$$\mathbb{E} \sqrt{\frac{1 - Z}{Z}} \leq 3\sqrt{2T}. \quad (2.5)$$

On $\{Z \geq 1/2\}$, the integrand is at most $\sqrt{2(1 - Z)}$, whose expectation is at most $\sqrt{2T}$ by Cauchy-Schwarz. If $q = \mathbb{P}(Z < 1/2)$, then $q \leq 2T$, and quantile domination gives

$$\mathbb{E}[Z^{-1/2} \mathbf{1}_{\{Z < 1/2\}}] \leq \int_0^q u^{-1/2} du = 2\sqrt{q} \leq 2\sqrt{2T}.$$

Here the quantile function of Z is at least its argument, and the event $Z < 1/2$ corresponds to its lowest q quantiles. This also justifies the estimate for laws with atoms.

Write $F_X(a) = \mathbb{P}(X \leq a)$ and $T_X(u) = \mathbb{P}(X' > X + u)$ for $u > 0$. Since $F_X(X)$ stochastically dominates a uniform variable, the same holds for the larger variable $F_X(X + u)$. Its expected complement is $\mathbb{P}(X' > X + u) = T_X(u)$, so

$$\mathbb{E} \sqrt{\frac{1 - F_X(X + u)}{F_X(X + u)}} \leq 3\sqrt{2T_X(u)}. \quad (2.6)$$

The reflected estimate follows from $1 - F_Y(Y - u) \geq \mathbb{P}(Y' \geq Y \mid Y)$ for $u > 0$; the latter variable is the distribution function of $-Y$ evaluated at $-Y$ and therefore also dominates uniform. The expected complement of $1 - F_Y(Y - u)$ is $\mathbb{P}(Y' \leq$

$Y - u) = T_Y(u)$ outside the finite set of positive differences between atoms. Thus the analogous estimate holds for $F_Y(Y - u)/(1 - F_Y(Y - u))$ for almost every $u > 0$.

Given $S = X + Y = s$, predict $\{X \leq a\}$ with probability

$$q_s(a) = \frac{F_X(a)(1 - F_Y(s - a))}{F_X(a)(1 - F_Y(s - a)) + (1 - F_X(a))F_Y(s - a)}.$$

Values at zero denominators may be assigned arbitrarily; the actual event has positive predicted probability outside irrelevant endpoints. For each fixed a , nonnegativity of binary relative entropy gives

$$H(\mathbf{1}_{\{X \leq a\}} \mid S) \leq \mathbb{E}[-\mathbf{1}_{\{X \leq a\}} \log q_S(a) - \mathbf{1}_{\{X > a\}} \log(1 - q_S(a))].$$

Integrate this nonnegative logarithmic loss in a and use Tonelli. For each realization (x, y) , split the integral at $a = x$. On $a > x$, write $a = x + u$ with $u > 0$. The loss becomes

$$\log \left(1 + \frac{1 - F_X(x + u)}{F_X(x + u)} \frac{F_Y(y - u)}{1 - F_Y(y - u)} \right).$$

Using $\log(1 + z) \leq \sqrt{z}$, independence, and (2.6), its expectation is at most

$$18\sqrt{T_X(u)T_Y(u)} \leq 9(T_X(u) + T_Y(u)).$$

On $a < x$, write $a = x - u$. The loss is

$$\log \left(1 + \frac{F_X(x - u)}{1 - F_X(x - u)} \frac{1 - F_Y(y + u)}{F_Y(y + u)} \right).$$

The reflected estimates give the same bound. Since

$$\int_0^\infty T_X(u) du = \frac{1}{2}\mathcal{D}(X),$$

integration over both halves proves (2.4). $\square$

To reveal a finite law gradually, place Poisson cuts on $\mathbb{R} \times (0, \infty)$ with intensity $da dt$. At time t , the spatial cuts have intensity t , with exponential gaps of mean $1/t$ between consecutive cuts. As t increases the partition refines, and each new cut supplies one binary label. Let $Q_t Y$ denote the cell containing Y . The cuts are independent of all input variables and are always included in the conditioning. Expressions such as $\mathbb{E}_Q \text{Var}(Y \mid Q_t Y)$ average both over the partitions and over the input cell. Only cuts separating support points affect a finite input's cell label.

The next lemma is the passage from absolute distances to variances. It makes this comparison after integrating over the refinement time. The resulting variance integral is exactly the logarithmic-scale integral of V_Y .

Lemma 2.3 (Poisson dispersion). *For every finite real variable Y ,*

$$\begin{aligned} \int_0^\infty \mathbb{E}_Q \mathcal{D}(Y \mid Q_t Y) dt &\leq 2 \int_0^\infty t \mathbb{E}_Q \text{Var}(Y \mid Q_t Y) dt \\ &= 12 \int_0^\infty V_Y(r) \frac{dr}{r} = 12\mathcal{E}(Y). \end{aligned} \quad (2.7)$$

Proof. Let $\rho = \mathcal{L}(Y)$. At intensity t , cells with left endpoint a and length r have intensity $t^2 e^{-tr} da dr$: left endpoints have intensity $t da$, and the distance to the next

cut is exponential of rate t . Tonelli's theorem and $\int_0^\infty t^3 e^{-tr} dt = 6r^{-4}$ therefore give

$$\begin{aligned} \int_0^\infty t \mathbb{E}_Q \text{Var}(Y \mid Q_t Y) dt &= \int_0^\infty \int_0^\infty \int_{\mathbb{R}} t^3 e^{-tr} F_\rho(I(a, r)) da dr dt \\ &= 6 \int_0^\infty V_\rho(r) \frac{dr}{r}. \end{aligned}$$

It remains to compare dispersion with variance. Fix atoms $x < y$ of masses p_x, p_y and put $d = y - x$. They share a cell with probability e^{-td} . Conditional on this event, its endpoints are $x - E_1/t$ and $y + E_2/t$, where E_1, E_2 are independent rate-one exponentials. Set

$$w(t) = \mathbb{E} [\rho([x - E_1/t, y + E_2/t))^{-1}].$$

The pair contributes $2p_x p_y d e^{-td} w(t)$ to the expected conditional dispersion and $p_x p_y d^2 e^{-td} w(t)$ to the expected conditional variance. Coupling the intervals with fixed E_1, E_2 shows that w is nondecreasing; it is bounded by $(p_x + p_y)^{-1}$.

For independent exponentials T, T' of rate d ,

$$\mathbb{E}[Tw(T)] - \mathbb{E}T \mathbb{E}w(T) = \frac{1}{2} \mathbb{E}[(T - T')(w(T) - w(T'))] \geq 0.$$

Since $\mathbb{E}T = 1/d$, this is exactly

$$\int_0^\infty e^{-td} w(t) dt \leq d \int_0^\infty t e^{-td} w(t) dt.$$

Multiplying by $2p_x p_y d$ and summing over unordered pairs gives (2.7). $\square$

Proposition 2.4 (Mixed addition). *For independent finite real variables $X_1, \dots, X_k$,*

$$\sum_{i=1}^k H(X_i) - H\left(\sum_{i=1}^k X_i\right) \leq 108k \sum_{i=1}^k \mathcal{E}(X_i). \quad (2.8)$$

Proof. For each input take an independent Poisson partition Q_t^i , with the cut realizations included in the conditioning. Set $S = \sum_i X_i$ and

$$J(t) = \mathbb{E}_Q H(Q_t^1 X_1, \dots, Q_t^k X_k \mid S).$$

At $t = 0$ there is one cell, so $J(0) = 0$. Every gap between consecutive atoms eventually receives a cut; hence the partitions distinguish all atoms almost surely. The entropies are bounded by $\sum_i H(X_i)$, and dominated convergence gives

$$J(\infty) = H(X_1, \dots, X_k \mid S) = \sum_i H(X_i) - H(S).$$

A new cut in coordinate i at a adds the conditional entropy of $\mathbf{1}_{\{X_i \leq a\}}$, given S and all previous cell labels. Consequently

$$J'(t) = \mathbb{E}_Q \sum_{i=1}^k \sum_C \mathbb{P}(X \in C) \int_{C_i} H(\mathbf{1}_{\{X_i \leq a\}} \mid S, X \in C) da, \quad (2.9)$$

where $C = C_1 \times \dots \times C_k$ ranges over tuples of cells and $X = (X_1, \dots, X_k)$. Terms of zero probability are omitted. To justify the derivative, restrict cuts to a compact interval containing all supports. The restricted process has finite rate. In time dt , one cut in da has probability $da dt + o(dt)$, and the probability of two or more cuts is $O(dt^2)$. The entropy change is uniformly bounded by $\sum_i H(X_i)$, so the contribution of multiple cuts, divided by dt , tends to zero. The chain rule for a

single cut therefore proves (2.9). Cuts outside the compact interval do not separate any support points and only relabel cells. Equivalently, each gap of length d between consecutive support points acquires its first cut after an independent exponential waiting time of rate d .

Fix a cell tuple C and condition on $X \in C$. Its coordinates remain independent, because C is a product of coordinate events. Apply Lemma 2.2 to the conditional law of X_i and the sum of the other conditional inputs. The indicator is constant outside C_i , so extending its threshold integral to $\mathbb{R}$ changes nothing. Independent-copy coupling and the triangle inequality give

$$\mathcal{D}\left(\sum_{j \neq i} X_j \mid X \in C\right) \leq \sum_{j \neq i} \mathcal{D}(X_j \mid X_j \in C_j).$$

Sum over i and average over the cell tuple. Each marginal conditional dispersion occurs at most k times. Thus

$$J'(t) \leq 9k \sum_i \mathbb{E}_Q \mathcal{D}(X_i \mid Q_i^t X_i). \quad (2.10)$$

The expectation on the right includes the probability of each cell.

Integrating (2.10) and applying Lemma 2.3 proves the proposition. $\square$

3. FINITE PRODUCTS AT A PRESCRIBED SCALE

Throughout the rest of the proof, Z is standard normal and

$$\mathcal{H}(Y; R) = I(Y; Y + RZ).$$

There is an absolute constant C_0 such that, for every bounded Y ,

$$|H(\lfloor Y/R \rfloor) - \mathcal{H}(Y; R)| \leq C_0. \quad (3.1)$$

The appendix proves this by bounding the integer rounding error of Gaussian noise. No smoothness of the law of Y is needed. The next estimate controls the information about Y left below resolution R . Its variance term adds $V_Y(r)$ over $0 < r < R$; the other main term comes from counting sums of independent copies.

Proposition 3.1 (Finite products). *Suppose $Y = \sum_{j=1}^L a_j D_j$, where the digits are independent and each has at most M values. Their laws need not be identical. For every positive integer k and every $R > 0$,*

$$H(Y) \leq \mathcal{H}(Y; R) + \frac{L(M-1) \log(k+1)}{k} + 108k \mathcal{E}_R(Y) + \frac{27}{2}k + C_0. \quad (3.2)$$

Proof. Let $Y_1, \dots, Y_k$ be independent copies of Y , and write

$$Q = \lfloor Y/R \rfloor, \quad Q_i = \lfloor Y_i/R \rfloor, \quad S = \sum_{i=1}^k Y_i.$$

Given $(Q_1, \dots, Q_k)$, the inputs remain independent. Apply Proposition 2.4 to their conditional laws and average. Since conditioning decreases the discrete entropy of S ,

$$kH(Y \mid Q) \leq H(S) + 108k^2 \mathbb{E}_Q \mathcal{E}(Y \mid Q).$$

At each scale, concavity gives $\mathbb{E}_Q V_{Y \mid Q}(r) \leq V_Y(r)$, where the left side evaluates V on each conditional law and then averages. Integrating over $0 < r < R$ bounds the

conditional energy below R by $\mathcal{E}_R(Y)$. Each cell has diameter at most R , so (2.3) bounds its remaining energy by $1/8$. Consequently

$$\mathbb{E}_Q \mathcal{E}(Y \mid Q) \leq \mathcal{E}_R(Y) + \frac{1}{8}.$$

Together with $H(Y) = H(Q) + H(Y \mid Q)$ and (3.1), this gives

$$H(Y) \leq \mathcal{H}(Y; R) + \frac{H(S)}{k} + 108k\mathcal{E}_R(Y) + \frac{27}{2}k + C_0.$$

To bound $H(S)$, realize the copies using independent digits $D_{i,j}$. For each j , the first $M-1$ multiplicities determine the sum of its k digits, since the last multiplicity is their complement in k . Each multiplicity lies in $\{0, \dots, k\}$. Thus

$$H(S) \leq \sum_{j=1}^L H\left(\sum_{i=1}^k D_{i,j}\right) \leq L(M-1) \log(k+1), \quad (3.3)$$

which proves (3.2). $\square$

4. GAUSSIAN DETECTION OF ORDERED MEANS

We use data processing and Pinsker's inequality (see [CT06]), and de Bruijn's identity and the score formula for Gaussian convolution (see [Sta59]). Their required forms are proved in the appendix. We write γ_t for the law of tZ ; the subscript denotes standard deviation, not variance.

We use the canonical realization

$$X = \sum_{j=1}^{\infty} t_{\omega_j} \prod_{\ell < j} r_{\omega_\ell}.$$

The series converges absolutely, since the family is finite and $\max_i |r_i| < 1$, and its law is ν . For every n it satisfies

$$X = B_n + R_n X^{(n)}, \quad (4.1)$$

where $X^{(n)}$ is independent of the first n letters and has law ν . All information about prefixes below refers to this coupling.

The first-letter identity is also the Bernoulli similarity case of the projection-entropy dimension formula of Feng–Hu [FH09, Theorem 2.8].

Lemma 4.1 (First-letter information). *Let X be the stationary variable coded by the independent letters, let $\alpha = \dim \nu$, and put $I_1(t) = I(\omega_1; X + tZ)$. Then*

$$I_1(t) = \mathcal{H}(X; t) - \sum_i p_i \mathcal{H}(X; t/r_i) \leq \alpha \chi_\mu \quad (t > 0), \quad (4.2)$$

and $I_1(t) \rightarrow \alpha \chi_\mu$ as $t \downarrow 0$.

Proof. Conditional on $\omega_1 = i$, the output has the law of $t_i + r_i X' + tZ$, with X' an independent copy of X . Gaussian symmetry, including when $r_i < 0$, gives its differential entropy as $\mathcal{H}(X; t/r_i) + h(tZ)$. Subtracting this conditional entropy from the output entropy gives the identity in (4.2).

Exact dimensionality and compact support give $\mathcal{H}(X; t) = \alpha \log(1/t) + o(\log(1/t))$ as $t \downarrow 0$; the comparison with quantized entropy is proved in the appendix. Set

$g(s) = \mathcal{H}(X; e^{-s})$ and $\ell_i = \log |r_i|^{-1}$. For $A > \max_i \ell_i$, a change of variables gives

$$\begin{aligned} \int_0^A I_1(e^{-s}) ds &= \sum_i p_i \left(\int_{A-\ell_i}^A g(s) ds - \int_{-\ell_i}^0 g(s) ds \right) \\ &= \alpha \chi_\mu A + o(A). \end{aligned}$$

Gaussian degradation shows that $I_1(e^{-s})$ is nondecreasing in s , and it is bounded by $H(p)$. Its limit must equal the limit of the displayed averages, namely $\alpha \chi_\mu$. Monotonicity gives the bound for every $t > 0$. $\square$

Lemma 4.2 (Uniform roughness). *If $\alpha = \dim \nu < 1$, there is $0 < \kappa \leq 1$ such that*

$$t \|(\nu * \gamma_t)'\|_1 \geq \kappa \quad (t > 0). \quad (4.3)$$

The law of $aX + c$, for every $a \neq 0$ and $c \in \mathbb{R}$, satisfies the same bound with the same κ .

Proof. Write $f_t = \nu * \gamma_t$ and $J(f) = \int (f')^2 / f$. Since $\frac{d}{dt} \mathcal{H}(X; t) = tJ(f_t) - t^{-1}$, de Bruijn's identity and Lemma 4.1 give

$$\sum_i p_i \int_t^{t/|r_i|} uJ(f_u) du = \chi_\mu - I_1(t) \geq (1 - \alpha)\chi_\mu.$$

Fisher information decreases under further Gaussian convolution, so

$$(1 - \alpha)\chi_\mu \leq \sum_i p_i \int_t^{t/|r_i|} uJ(f_u) du \leq \frac{t^2}{2} J(f_t) \sum_i p_i (|r_i|^{-2} - 1).$$

Thus $t^2 J(f_t) \geq c_0 > 0$ uniformly, where one may take

$$c_0 = \frac{2(1 - \alpha)\chi_\mu}{\sum_i p_i (|r_i|^{-2} - 1)}.$$

For the score $S_t = f'_t(X + tZ)/f_t(X + tZ)$, the score identity gives $\mathbb{E}|S_t|^4 \leq 3t^{-4}$, while $\mathbb{E}S_t^2 = J(f_t)$. Hölder therefore yields

$$t \|f'_t\|_1 = t \mathbb{E}|S_t| \geq t \frac{(\mathbb{E}S_t^2)^{3/2}}{(\mathbb{E}S_t^4)^{1/2}} \geq \frac{c_0^{3/2}}{\sqrt{3}}.$$

Choose $\kappa = \min\{1, c_0^{3/2}/\sqrt{3}\}$. For the law of $aX + c$, a change of variables replaces t by $t/|a|$ in the quantity $t \|f'_t\|_1$, proving the last assertion. $\square$

To see which variance a cut reveals, let a binary label A divide the law of U into two parts of probabilities q, p and means m_0, m_1 . The law of total variance reads

$$\text{Var}(U) = q \text{Var}(U \mid A = 0) + p \text{Var}(U \mid A = 1) + pq(m_1 - m_0)^2. \quad (4.4)$$

The final term is the variance of the conditional mean: it is the part revealed by the label. The next lemma shows that, for ordered parts in an interval of length r , the Gaussian observation detects this contribution after normalization by r^2 .

Lemma 4.3 (Ordered detection). *Let (A, U) be independent of X , where A is binary and U lies in an interval of length r . Suppose that the two conditional laws of U given A are stochastically ordered, with probabilities q, p and mean gap $d \geq 0$. Assume X satisfies (4.3). At $s = 4r/\kappa$,*

$$I(A; X + U + sZ) \geq \frac{\kappa^4 p q d^2}{128 r^2}. \quad (4.5)$$

Proof. The case $pq = 0$ is immediate. Otherwise translate so that $U \in [0, r]$, and couple its two conditional laws by their quantiles. The resulting variables satisfy $U_0 \leq U_1$ and $\mathbb{E}(U_1 - U_0) = d$. The density f_s of $X + sZ$ satisfies $\|f_s''\|_1 \leq 2s^{-2}$. If g_0, g_1 are the two conditional output densities, independence of X from (A, U) gives

$$g_1 - g_0 = -\mathbb{E} \int_{U_0}^{U_1} f_s'(\cdot - u) du.$$

Approximate this difference by $-df_s'$. The translation estimate $\|f_s'(\cdot - u) - f_s'\|_1 \leq u \|f_s''\|_1$ for $u \in [0, r]$ then gives

$$g_1 - g_0 = -df_s' + E_s, \quad \|E_s\|_1 \leq 2rd/s^2.$$

In particular,

$$\|g_1 - g_0\|_1 \geq d\kappa/s - 2rd/s^2 = \kappa^2 d/(8r).$$

Pinsker's inequality for the mixture $qg_0 + pg_1$ gives

$$I(A; X + U + sZ) \geq \frac{1}{2}pq \|g_1 - g_0\|_1^2,$$

and (4.5) follows. $\square$

We now sum the information from successive cuts. The variance revealed at each cut is given by (4.4). Averaging over the Poisson partitions will turn their normalized contributions into the integral of V_Y below R .

Proposition 4.4 (Detection by Poisson cuts). *Let Y be finite and independent of a bounded X satisfying (4.3). For every $R > 0$,*

$$c_\kappa \int_0^R V_Y(r) \frac{dr}{r} \leq \lim_{t \downarrow 0} I(Y; X + Y + tZ) - I(Y; X + Y + RZ) + \frac{1}{4}. \quad (4.6)$$

Here $c_\kappa = \kappa^4/(128e)$, and the integral is $\mathcal{E}_R(Y)$. The limit exists by Gaussian degradation and is at most $H(Y)$.

Proof. There is nothing to prove when Y is a point mass. Otherwise let $\delta > 0$ be its smallest positive atom spacing, and fix $0 < t < \min\{R, \delta\}$. Write $O_a = X + Y + aZ$ and $\rho = \mathcal{L}(Y)$. Use the Poisson cuts from Proposition 2.4. Let K be their partition at intensity $1/R$. We always include the independent cut realization in the conditioning.

The chain rule separates the information about $K(Y)$ from the information within its cell. The former increases as noise decreases. For the latter, Gaussian capacity gives the averaged bound

$$I(Y; O_t) - I(Y; O_R) \geq \mathbb{E}_K I(Y; O_t | K(Y)) - \frac{1}{4}. \quad (4.7)$$

Indeed, dropping the independent background bounds the coarse conditional information by $\mathbb{E}_K \text{Var}(Y | K(Y))/(2R^2)$. The conditional mean minimizes squared error, so this variance is at most the expected squared distance from Y to the cell midpoint. For every fixed $Y = y$, the distances from y to the cuts on its left and right are independent exponentials of mean R . Thus the expected squared distance to the midpoint is $R^2/2$. The averaged coarse information is therefore at most $(R^2/2)/(2R^2) = 1/4$.

Continue adding cuts after intensity $1/R$. The chain rule decomposes $\mathbb{E}_K I(Y; O_t | K(Y))$ into the expected information contributions of the subsequent binary labels, weighted by their parent probabilities. Only finitely many cuts separate distinct atoms, and eventually every atom is distinguished. Retain only cuts with parent length $r \leq R$. Write I for such a parent and p, q, d for its conditional child masses

and mean gap. If both children have positive mass, then $r \geq \delta > t$. The background is still independent after conditioning on $Y \in I$, and the child laws are ordered. Degrade the noise to $4r/\kappa$ and apply Lemma 4.3. This cut contributes at least

$$\frac{\kappa^4}{128} \rho(I) \frac{pqd^2}{r^2}.$$

The minimum spacing only chooses t for this fixed finite law; no bound on that spacing is required.

It remains to calculate the expected variance sum. Put $A(r) = r^3 V_\rho(r) = \int_{\mathbb{R}} F_\rho(I(a, r)) da$. For a cut at $a + u$ in $I(a, r)$, (4.4) gives

$$\rho(I(a, r)) pqd^2 = F_\rho(I(a, r)) - F_\rho(I(a, u)) - F_\rho(I(a + u, r - u)).$$

Integrating over $a \in \mathbb{R}$ and $0 < u < r$ yields $rA(r) - 2 \int_0^r A(u) du$. At intensity τ , parent intervals have intensity $\tau^2 e^{-\tau r} da dr$, and new cuts have intensity $du d\tau$. For $r \leq R$,

$$\int_{1/R}^{\infty} \tau^2 e^{-\tau r} d\tau = \frac{2}{r^3} e^{-r/R} \left(1 + \frac{r}{R} + \frac{r^2}{2R^2} \right) \geq \frac{2}{er^3}.$$

Tonelli applies because the cut costs are nonnegative. The expected normalized variance sum is therefore at least B_R/e , where

$$\begin{aligned} B_R &= 2 \int_0^R \frac{rA(r) - 2 \int_0^r A(u) du}{r^5} dr \\ &= \mathcal{E}_R(Y) + \frac{1}{R^4} \int_0^R A(u) du \geq \mathcal{E}_R(Y). \end{aligned} \quad (4.8)$$

The equality follows by interchanging the two integrals and using $\int_u^R r^{-5} dr = (u^{-4} - R^{-4})/4$. The integrals are finite, since A vanishes below δ . Combining this computation with (4.7) proves the bound for every $0 < t < \min\{R, \delta\}$. Let $t \downarrow 0$. $\square$

5. CONDITIONAL PRODUCTS OF WORD MAPS

For each n and $R > 0$, define

$$Q_n(R) = \mathbb{E}_{R_n} \mathcal{E}_R(B_n | R_n).$$

Equivalently,

$$Q_n(R) = \int_0^R \mathbb{E}_{R_n} V_{B_n | R_n}(r) \frac{dr}{r}.$$

At each scale the local variance is evaluated on the conditional translation law for each ratio, and then averaged. Thus $Q_n(R)$ accumulates the normalized variance still present below scale R . The following comparison is finite and makes no assumption on $\dim \nu$.

Proposition 5.1 (Finite entropy comparison). *For positive integers n, k and $R > 0$, with $a_k = \frac{27}{2}k + C_0$, one has*

$$H(R_n, B_n) \leq nH(R_1) + \mathcal{H}(X; R) + \frac{n(m-1) \log(k+1)}{k} + 108kQ_n(R) + a_k. \quad (5.1)$$

Proof. Condition on the individual ratios $C = (r_{\omega_1}, \dots, r_{\omega_n})$. Since R_n is a function of C ,

$$H(R_n, B_n) \leq H(C, B_n) = nH(R_1) + H(B_n | C). \quad (5.2)$$

For a fixed $C = c$ of positive probability, the translations $D_j = t_{\omega_j}$ are independent, each with at most m values, and

$$B_n = D_1 + c_1 D_2 + \cdots + (c_1 \cdots c_{n-1}) D_n.$$

Apply Proposition 3.1 to each fixed conditional law, with $L = n$ and $M = m$, and then average in c . Thus

$$H(B_n | C) \leq \mathbb{E}_C \mathcal{H}(B_n | C; R) + \frac{n(m-1) \log(k+1)}{k} + 108k \mathbb{E}_C \mathcal{E}_R(B_n | C) + a_k.$$

Write $X = B_n + R_n X'$ with an independent stationary tail X' . Given $C = c$, the ratio is fixed and the tail is independent of B_n . Entropy increases under independent convolution, so

$$h(B_n + RZ | C = c) \leq h(B_n + R_n X' + RZ | C = c) = h(X + RZ | C = c).$$

Averaging and using $h(X + RZ | C) \leq h(X + RZ)$, then subtracting $h(RZ)$, gives $\mathbb{E}_C \mathcal{H}(B_n | C; R) \leq \mathcal{H}(X; R)$. Finally R_n is a function of C , so concavity gives

$$\mathbb{E}_C \mathcal{E}_R(B_n | C) \leq \mathbb{E}_{R_n} \mathcal{E}_R(B_n | R_n) = Q_n(R).$$

Substitute these bounds into (5.2). $\square$

6. SATURATION OF PREFIX INFORMATION

Let $W_n = (\omega_1, \dots, \omega_n)$ and $L_n = -\log |R_n|$. All information uses the canonical coupling (4.1).

Lemma 6.1 (Prefix saturation). *For every $t > 0$ and $n \geq 1$,*

$$I(W_n; X + tZ) \leq n\alpha\chi_\mu. \quad (6.1)$$

For every $b > \chi_\mu$,

$$I(W_n; X + e^{-bn} Z) = n\alpha\chi_\mu + o(n). \quad (6.2)$$

Proof. Let $I_1(t) = I(\omega_1; X + tZ)$. Given W_{j-1} , subtract B_{j-1} from the observation and divide by R_{j-1} . The remaining tail and its first letter retain their joint law, and Gaussian symmetry gives noise standard deviation $t/|R_{j-1}|$ for either sign of R_{j-1} . The chain rule gives, with $R_0 = 1$,

$$I(W_n; X + tZ) = \sum_{j=1}^n \mathbb{E} I_1(t/|R_{j-1}|) \leq n\alpha\chi_\mu$$

by Lemma 4.1. Every map contracts, so $|R_{j-1}| \geq |R_n|$. Since I_1 decreases with its noise parameter, the same identity gives

$$\mathbb{E} I_1(t/|R_n|) \leq \frac{1}{n} I(W_n; X + tZ) \leq \alpha\chi_\mu.$$

For $t = e^{-bn}$, the law of large numbers gives $t/|R_n| = e^{-bn+L_n} \rightarrow 0$ almost surely. Lemma 4.1 and bounded convergence make the left side tend to $\alpha\chi_\mu$, proving (6.2). $\square$

Proposition 6.2 (Sublinear fine variance). *If $\alpha < 1$, then for every $b > \chi_\mu$,*

$$Q_n(e^{-bn}) = o(n). \quad (6.3)$$

Proof. Put $\rho_n = e^{-bn}$ and $O_a = X + aZ$. The conditional output given W_n depends only on its map, so

$$I(W_n; O_a) = I(R_n; O_a) + I(B_n; O_a | R_n).$$

For $t < \rho_n$, the difference of the first terms at $a = t$ and $a = \rho_n$ is nonnegative by Gaussian degradation. Given $R_n = s$, the remaining channel is $B_n + sX' + aZ$. Its background sX' is independent of B_n and has the same roughness constant as ν . Apply Proposition 4.4 and average over the finitely many values of s . Thus

$$\begin{aligned} c_\kappa Q_n(\rho_n) &\leq \lim_{t \downarrow 0} [I(W_n; O_t) - I(W_n; O_{\rho_n})] + \frac{1}{4} \\ &\leq n\alpha\chi_\mu - I(W_n; O_{\rho_n}) + \frac{1}{4} = o(n), \end{aligned}$$

by Lemma 6.1. The error is paid once after averaging. The limit in t is taken for each fixed n , so no uniform spacing estimate is needed. $\square$

7. PROOF OF THE DIMENSION FORMULA

Proof of Theorem 1.1. Suppose first that $\alpha < 1$. Fix $b > \chi_\mu$ and a positive integer k , apply (5.1) with $R = e^{-bn}$, and let $n \rightarrow \infty$. Exact dimensionality gives $\mathcal{H}(X; e^{-bn})/n \rightarrow \alpha b$, while Proposition 6.2 gives $Q_n(e^{-bn})/n \rightarrow 0$. Thus

$$h_\mu \leq H(R_1) + \alpha b + \frac{(m-1)\log(k+1)}{k}.$$

Let $k \rightarrow \infty$ and then $b \downarrow \chi_\mu$ to obtain

$$h_\mu \leq H(R_1) + \alpha\chi_\mu. \quad (7.1)$$

Now fix N and regard the distinct N -letter maps, with their induced probabilities, as a new system. Its stationary measure is still ν , its Lyapunov exponent is $N\chi_\mu$, and its entropy rate is

$$h_{\mu^{*N}} = \lim_{q \rightarrow \infty} \frac{H(\mu^{*Nq})}{q} = Nh_\mu.$$

Its single-step contraction is R_N . Applying (7.1) to this system and dividing by N gives

$$h_\mu \leq \frac{H(R_N)}{N} + \alpha\chi_\mu.$$

If N_i counts occurrences of letter i , then $R_N = \prod_i r_i^{N_i}$, including its sign. Hence $H(R_N) \leq m \log(N+1) = o(N)$. Let $N \rightarrow \infty$. This proves $h_\mu \leq \alpha\chi_\mu$ whenever $\alpha < 1$. No uniform estimate in N is needed: (7.1) is applied separately to each fixed iterated system.

For the opposite inequality, regard all distinct n -letter similarities as a new alphabet, merging coincident maps and adding their probabilities. Its Shannon entropy is $H(R_n, B_n)$ and its mean logarithmic contraction is $n\chi_\mu$. The usual coding upper bound gives $\dim \nu \leq \frac{H(R_n, B_n)}{n\chi_\mu}$. Here is the coding argument. The point-mass case is immediate, so let the support have diameter $D > 0$. The law of large numbers for successive letters of the new alphabet shows that a typical symbolic cylinder of length q has probability $\exp(-qH(R_n, B_n) + o(q))$ and its geometric image has diameter $D \exp(-qn\chi_\mu + o(q))$. This image contains the coded point x , and hence lies in the ball about x of that diameter. The measure of the ball is at least the symbolic cylinder probability, even when other cylinders overlap it. The lower local dimension along these radii is therefore at most $\frac{H(R_n, B_n)}{n\chi_\mu}$. Exact dimensionality

identifies it with $\dim \nu$ almost everywhere. Take the infimum over n and also use $\dim \nu \leq 1$. Together with the preceding paragraph this proves the formula; when $\alpha = 1$, the coding upper bound already implies $h_\mu \geq \chi_\mu$. $\square$

APPENDIX A. ENTROPY AND GAUSSIAN PRELIMINARIES

We recall the short analytic facts used in the argument. Entropy chain rules, nonnegativity of relative entropy, and Jensen's inequality are used in their usual elementary forms.

Exact dimensionality and quantization. Suppose a compactly supported law ρ is exact dimensional of dimension α . For the mesh- r cell $Q_r(x)$ containing x , $\rho(Q_r(x)) \leq \rho([x-r, x+r])$. Fatou's lemma therefore gives

$$\liminf_{r \downarrow 0} \frac{H(\lfloor X/r \rfloor)}{\log(1/r)} \geq \alpha.$$

For the reverse bound, given $\epsilon > 0$, choose a set E with $\rho(E) > 1 - \epsilon$ and $r_0 > 0$ such that $\rho([x-r, x+r]) \geq r^{\alpha+\epsilon}$ whenever $x \in E$ and $r < r_0$. Choose one point of E in each mesh cell meeting E . The radius- r intervals about these points have bounded overlap, so there are at most $4r^{-\alpha-\epsilon}$ such cells. Compact support gives at most C/r cells in total. Splitting according to E consequently gives

$$H(\lfloor X/r \rfloor) \leq (\alpha + 2\epsilon) \log(1/r) + O_\epsilon(1).$$

Let $\epsilon \downarrow 0$.

To compare Gaussian information, write $X/r = K + U$ with K integral and $0 \leq U < 1$. The chain rule gives

$$I(X; X + rZ) = H(K) - H(K \mid K + U + Z) + I(U; U + Z \mid K).$$

The last term is at most $\frac{1}{2} \log(1 + 1/4)$ by the Gaussian capacity bound, applied conditionally on K . For the preceding conditional entropy, put $D = K - \lfloor K + U + Z \rfloor = -\lfloor U + Z \rfloor$. Given the output $K + U + Z$, knowing D determines K , so $H(K \mid K + U + Z) \leq H(D)$. Moreover $\mathbb{E}|D| \leq 1 + \mathbb{E}|Z|$. Comparing the law of D with the probability mass function $q_j = ce^{-|j|}$ on $\mathbb{Z}$ gives $H(D) \leq -\log c + \mathbb{E}|D| = O(1)$, uniformly in the joint law of K, U . It follows that $\mathcal{H}(X; r) = H(K) + O(1)$, which gives (3.1) and, for an exact-dimensional law, $\mathcal{H}(X; r) = \alpha \log(1/r) + o(\log(1/r))$. The capacity bound itself follows from nonnegativity of relative entropy against a Gaussian with matching mean and variance: $h(W) \leq \frac{1}{2} \log(2\pi e \operatorname{Var} W)$.

Pinsker and Gaussian smoothing. For densities f, g , let $A = \{f \geq g\}$. Passing to the indicator of A cannot increase relative entropy. The binary divergence, as a function of its first parameter, has second derivative $1/[p(1-p)] \geq 4$. Thus

$$D(f \| g) \geq 2 \left(\int_A (f - g) \right)^2 = \frac{1}{2} \|f - g\|_1^2.$$

Applied to a mixture $qg_0 + pg_1$, this yields $I(A; O) \geq pq \|g_1 - g_0\|_1^2 / 2$. Indeed, the distances of g_0, g_1 from the mixture are respectively $p \|g_1 - g_0\|_1$ and $q \|g_1 - g_0\|_1$, and $qp^2 + pq^2 = pq$.

If f_t is the density of $X + tZ$ for bounded X , direct differentiation under the Gaussian integral gives

$$\partial_t f_t = t f_t'', \quad \frac{f_t'(X + tZ)}{f_t(X + tZ)} = -t^{-1} \mathbb{E}[Z \mid X + tZ].$$

Since X is bounded, the Gaussian tails justify differentiation under the integral and integration by parts. Using $\int \partial_t f_t = 0$, we obtain

$$\frac{d}{dt} h(f_t) = -t \int f_t'' \log f_t = t \int \frac{(f_t')^2}{f_t} = tJ(f_t).$$

The score identity and conditional Jensen give the fourth moment bound $3t^{-4}$. If $g = f * \eta$, its score is the conditional expectation of the score of f given the sum. Jensen gives $J(g) \leq J(f)$. Finally $\|f_t''\|_1 \leq \|\gamma_t''\|_1 \leq 2t^{-2}$.

The interpolation step used above is

$$\mathbb{E}|S| \geq \frac{(\mathbb{E}S^2)^{3/2}}{(\mathbb{E}S^4)^{1/2}},$$

by Hölder's inequality. This completes the Gaussian estimates.

A bounded-increment tail estimate. For bounded ℓ with mean χ_μ and fixed $\delta > 0$, choose $\theta > 0$ with $\log \mathbb{E}e^{\theta(\ell - \chi_\mu)} < \theta\delta/2$. Independence and the exponential Markov inequality give $\mathbb{P}(\sum_{j=1}^n \ell_j > (\chi_\mu + \delta)n) \leq e^{-\theta\delta n/2}$. Apply the same bound to $-\ell$ and sum over n . Borel–Cantelli then gives the law of large numbers $L_n/n \rightarrow \chi_\mu$ used above.

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